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Interruption Success or Failure Caused by Arc Behavior in Air Switchgear

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Abstract

To clarify the factors leading to arc reignition in air switchgear, the spatiotemporal variations of arc behavior, temperature, and electrical conductivity, as well as hot gas emissions near the first current zero, were measured. In cases with short interruption times, the arc spot on the movable electrode tended to move away from the contact gap, and the electrical conductivity near the spot ranged from 10^{-1} to 10^2 S/m, approximately one order of magnitude lower than in unsuccessful cases. This suggests that the spot position immediately before the current zero is a critical factor for successful thermal interruption. In contrast, in cases where reignition occurred, the emission intensity of gas ejected from the arc-quenching chamber was weak, and the ratio of the total gas to the total arc path temperature was less than half that in successful interruptions. Since the injected electrical energy between the electrodes showed no significant difference between successful and unsuccessful cases, it is suggested that the strong gas emission contributes to suppressing arc path luminosity, thereby preventing reignition.

1 Section title

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Funding

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Data availability

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Supplementary data

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References